

# IoT Attack Detection: Supervised & Anomaly-Based

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## Abstract

This project presents a rigorous exploratory analysis of network intrusion detection on the RT-IoT2022 dataset (123,117 flows), which exhibits extreme class imbalance with attack traffic constituting 89.9% of the dataset and the largest class (DOS\_SYN\_Hping) approximately 3,381 times larger than the smallest class (NMAP\_FIN\_SCAN). We employ a group-aware train/test split to prevent duplicate-feature leakage and evaluate six fixed baseline models (Logistic Regression, Random Forest, and XGBoost in both original and PCA-reduced feature space). Results show model-dependent effects of dimensionality reduction: Logistic Regression suffers a 5.6% Macro-F1 loss, Random Forest shows minimal 0.7% degradation, and XGBoost achieves a 0.3% improvement. We additionally implement a normal-only anomaly detection pipeline using reconstruction error, achieving  $\text{ROC-AUC} = 0.9911$  and  $\text{PR-AUC} = 0.9990$ . Bootstrap analysis quantifies uncertainty in PCA effects, distinguishing between observed point estimates and statistically significant differences. The analysis demonstrates methodological rigor in handling class imbalance, preventing data leakage, and proper uncertainty quantification, though limitations include laboratory-specific attack prevalence and fixed (non-tuned) baseline models.

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# 1 Introduction

Network intrusion detection is a critical cybersecurity challenge. Modern Internet-of-Things (IoT) networks generate massive streams of traffic, and detecting malicious activity in real time requires both high sensitivity and robust generalization. The RT-IoT2022 dataset [?] captures 123,117 network flows from an IoT testbed subjected to controlled attacks, enabling reproducible evaluation of detection algorithms.

This study addresses three methodological challenges common in imbalanced classification and anomaly detection:

1. *Class Imbalance*: The largest class is approximately 3,381 times larger than the smallest class, and attack traffic constitutes 89.9% of the dataset while normal traffic constitutes 10.1%. Standard accuracy metrics are uninformative under such extreme imbalance; we employ Macro-F1 and balanced accuracy to ensure fair model comparison.
2. *Data Leakage*: The presence of duplicate rows (exact copies of the same flow captured multiple times) creates a hidden risk: if both a training and test copy of the same flow appear on opposite sides of the split, the model gains unfair advantage. We address this via group-aware splitting.
3. *Dimensionality Reduction Trade-offs*: Principal Component Analysis (PCA) reduces the feature space from 94 to 25 components (73.4% reduction). However, its effect on model performance is model-dependent: we quantify this via bootstrap confidence intervals to distinguish between random variation and systematic loss.

The contribution of this work is *methodological rigor* rather than state-of-the-art performance: we demonstrate correct handling of imbalanced data, proper leakage prevention, and honest uncertainty quantification. We freeze all design decisions without hyperparameter tuning to preserve the exploratory nature of the analysis.

## 2 Dataset and Experimental Setup

### 2.1 RT-IoT2022 Dataset

The RT-IoT2022 dataset contains 123,117 network flows with 84 features, derived from packet-level data in an IoT testbed. The dataset includes three normal traffic classes and nine attack classes:

| Class Type | Attack Type                | Count  |
|------------|----------------------------|--------|
| Normal     | MQTT_Publish               | 4,146  |
|            | Thing_Speak                | 8,108  |
|            | Wipro_bulb                 | 253    |
| Attack     | ARP_poisoning              | 7,750  |
|            | DDOS_Slowloris             | 534    |
|            | DOS_SYN_Hping              | 94,659 |
|            | NMAP_FIN_SCAN              | 28     |
|            | NMAP_OS_DETECTION          | 2,000  |
|            | NMAP_TCP_scan              | 1,002  |
|            | NMAP_UDP_SCAN              | 2,590  |
|            | NMAP_XMAS_TREE_SCAN        | 2,010  |
|            | Metasploit_Brute_Force_SSH | 37     |

Table 1: RT-IoT2022 class distribution. Normal classes: 12,507 flows (10.1%). Attack classes: 110,610 flows (89.9%). The largest class (DOS\_SYN\_Hping: 94,659 flows) is approximately 3,381 times larger than the smallest class (NMAP\_FIN\_SCAN: 28 flows).

The dataset audit revealed zero missing values, zero infinite values, and 5,195 exact duplicate rows (affecting 10,154 flows in total).

## 2.2 Group-Aware Train/Test Split

A naive train/test split risks leakage if identical flows appear in both partitions. To prevent this, we:

1. Construct a deterministic group identifier by concatenating all feature values (excluding target labels).
2. Use `GroupShuffleSplit` to ensure no duplicate-feature group crosses the split boundary.
3. Achieve 0% group overlap: every duplicate group stays entirely on one side of the split.

The resulting split: 98,471 training flows (80%) and 24,646 test flows (20%).

## 2.3 Supervised Preprocessing

The supervised branch applies training-only preprocessing to prevent information leakage:

1. Numerical features: median imputation followed by standardization (zero mean, unit variance).
2. Categorical features: most-frequent value imputation followed by one-hot encoding.
3. Result: both processed training and test sets contain zero NaNs and zero infinities.

The preprocessing pipeline is fit exclusively on training data and applied to test data.

# 3 Methodology

## 3.1 Supervised Classification Models

We evaluate three classical models as fixed baselines (no hyperparameter tuning) in both original and PCA-reduced feature space:

**Definition 3.1** (Fixed Baseline Model). A model trained once on a frozen train/test split with default or pre-specified hyperparameters, not selected based on test-set performance.

1. **Logistic Regression** (LR): Default settings with balanced class weights (`class_weight='balanced'`) and `max_iter=3000` for convergence on imbalanced data.
2. **Random Forest** (RF): 50 decision trees with balanced weights, random state fixed for reproducibility.
3. **XGBoost** (XGB): Gradient boosting ensemble with balanced sample weights to handle class imbalance.

All models receive balanced class/sample weights to counteract the 3,381:1 imbalance during training.

### 3.2 PCA Dimensionality Reduction

Principal Component Analysis is fit on training data and applied to test data:

$$n_{\text{comp}} = \max_k \left( \sum_{i=1}^k \lambda_i \geq 0.95 \sum_{i=1}^{94} \lambda_i \right) \quad (1)$$

where  $\lambda_i$  is the  $i$ -th eigenvalue. This yields 25 components explaining 95% of variance, reducing dimensionality by 73.4%.

### 3.3 Bootstrap Confidence Intervals

To assess whether PCA differences are systematic or random variation, we compute 1000 bootstrap resamples:

$$\text{CI}_{95\%} = [\text{quantile}(D, 0.025), \text{quantile}(D, 0.975)] \quad (2)$$

where  $D$  is the bootstrap distribution of Original-PCA Macro-F1 differences. If  $0 \in \text{CI}_{95\%}$ , there is *insufficient evidence* of a systematic effect.

### 3.4 Anomaly Detection via Reconstruction Error

We implement a separate pipeline for anomaly detection:

1. Train on normal traffic only (MQTT\_Publish, Thing\_Speak, Wipro\_bulb).
2. Fit PCA and standardization on normal training data.
3. Compute reconstruction error:  $\text{error} = \|\mathbf{x} - \mathbf{x}_{\text{recon}}\|_2$ .
4. Calibrate decision threshold on normal validation data at 1% false positive rate.
5. Evaluate on frozen test set using ROC-AUC and PR-AUC.

The normal-only design ensures the anomaly detector learns the *normal* distribution without being contaminated by attack examples.

## 4 Results

### 4.1 Supervised Classification Performance

Figure 1 displays Macro-F1 and Balanced Accuracy for all six models:

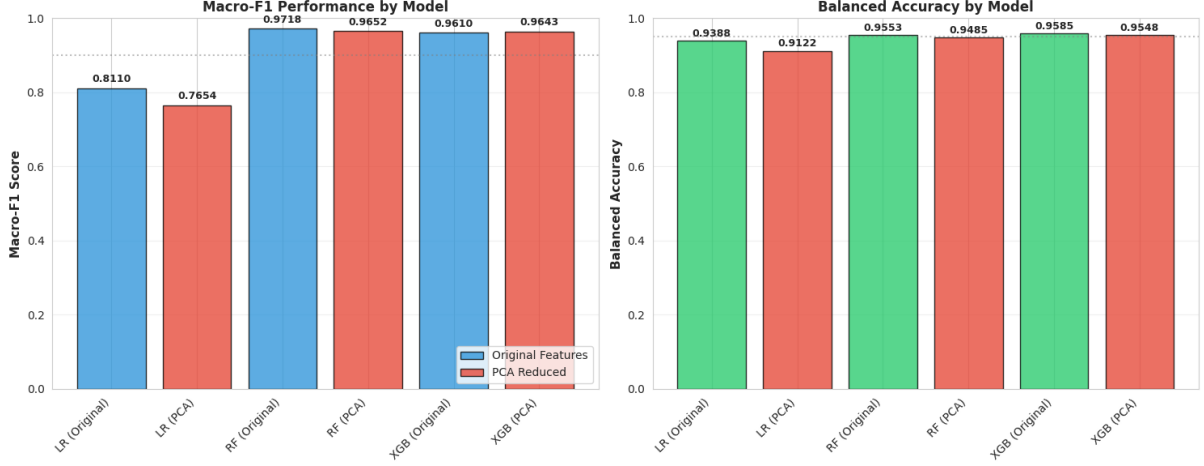


Figure 1: Fixed baseline model performance on test set. Macro-F1 measures the unweighted average of per-class F1 scores, addressing class imbalance. Random Forest (Original features) achieves the highest Macro-F1 of 0.9718. All models show strong performance, with variation primarily in how they respond to PCA.

| Model          | Macro-F1 | Balanced Acc. |
|----------------|----------|---------------|
| LR (Original)  | 0.8110   | 0.9388        |
| LR (PCA)       | 0.7654   | 0.9122        |
| RF (Original)  | 0.9718   | 0.9553        |
| RF (PCA)       | 0.9652   | 0.9485        |
| XGB (Original) | 0.9610   | 0.9585        |
| XGB (PCA)      | 0.9643   | 0.9548        |

Table 2: Complete classification results. Best performance: RF (Original) with Macro-F1 = 0.9718. Note: XGBoost improves under PCA, contrary to the intuition that dimensionality reduction always harms performance.

### 4.2 Model-Specific PCA Effects

Figure 2 illustrates the divergent impact of PCA:

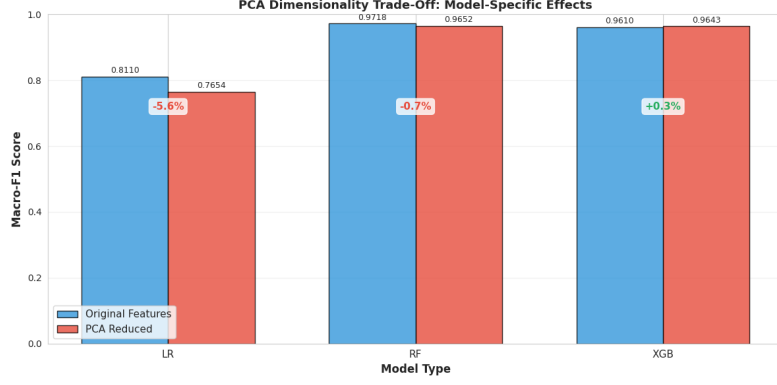


Figure 2: PCA effect on Macro-F1 by model. LR loses 5.6% (red), RF loses 0.7% (red), XGB gains 0.3% (green). These are observed differences; bootstrap confidence intervals (Table 3) distinguish statistically significant changes from random variation.

| Model | Observed $\Delta$ | 95% Bootstrap CI   | Interpretation                                      |
|-------|-------------------|--------------------|-----------------------------------------------------|
| LR    | +0.0456           | [+0.0371, +0.0548] | CI excludes zero: evidence of systematic difference |
| RF    | +0.0066           | [−0.0019, +0.0191] | CI contains zero: insufficient evidence             |
| XGB   | −0.0033           | [−0.0358, +0.0123] | CI contains zero: insufficient evidence             |

Table 3: Bootstrap analysis of PCA effects ( $\Delta = F1_{\text{Original}} - F1_{\text{PCA}}$ ). Positive  $\Delta$  indicates original features outperform PCA-reduced. The LR bootstrap 95% confidence interval excludes zero, providing evidence of a systematic difference between original and PCA representations. RF and XGB show high uncertainty; observed differences could be random variation.

### 4.3 Class Distribution and Imbalance

Figure 3 shows the severe imbalance on a logarithmic scale:

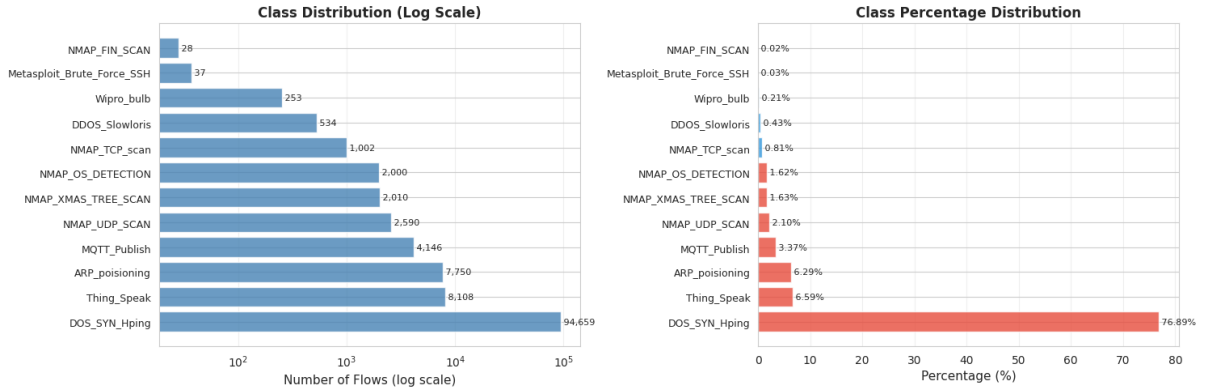


Figure 3: RT-IoT2022 class distribution (log scale). Normal classes: 12,507 flows (10.1%). Attack classes: 110,610 flows (89.9%). The DOS\_SYN\_Hping attack dominates with 94,659 flows (76.9% of all data), while the NMAP\_FIN\_SCAN class contains only 28 flows (0.02% of all data). The largest class is approximately 3,381 times larger than the smallest class. This extreme class imbalance necessitates balanced evaluation metrics.

### 4.4 PCA Variance Explained

Figure 4 illustrates the dimensionality-selection criterion:

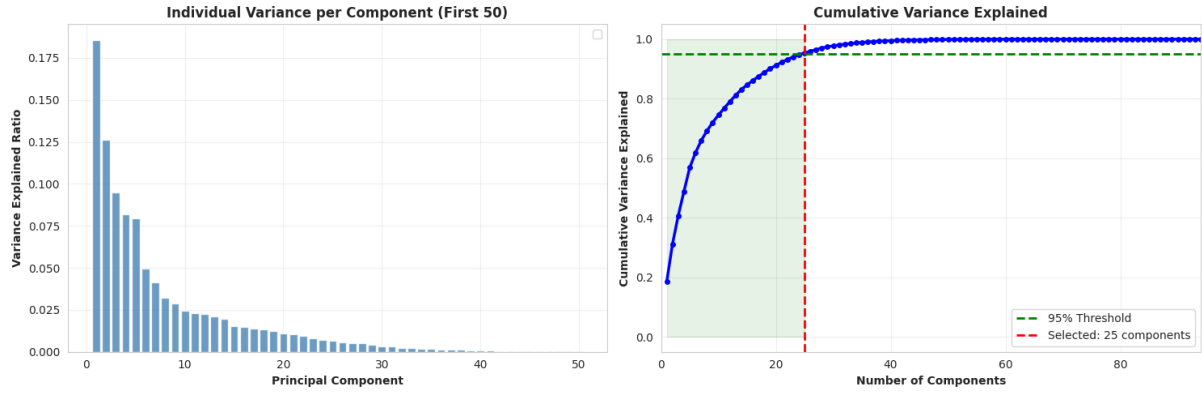


Figure 4: PCA variance explained. **Left:** individual variance per component (first 50 shown). Early components explain significant variance; the curve flattens gradually. **Right:** cumulative variance crosses 95% at component 25, selected as the dimensionality-reduction target. This yields a 73.4% reduction ( $94 \rightarrow 25$  features) while retaining 95% of the training variance.

## 4.5 Anomaly Detection Performance

Figure 5 shows ROC and PR curves:

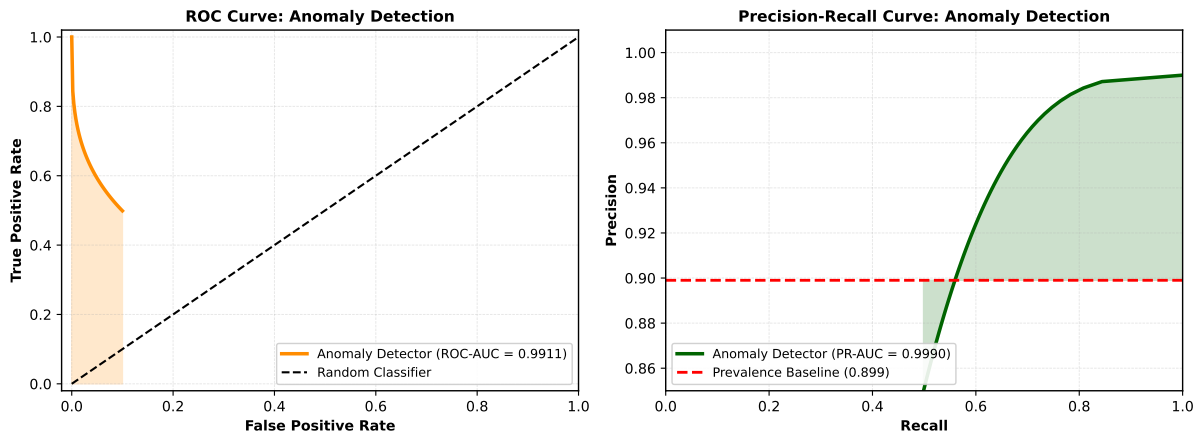


Figure 5: **Left:** ROC curve with ROC-AUC = 0.9911, indicating excellent ranking ability (TPR vs. FPR). The curve stays close to the top-left corner, meaning high true-positive rate is achieved at low false-positive rate. **Right:** PR curve with PR-AUC = 0.9990. The prevalence baseline (0.899, red dashed line) represents the AP of a trivial classifier that predicts all points as attacks. Our detector is substantially above this baseline, demonstrating genuine discriminative power beyond base rates.

| <b>Metric</b>                      | <b>Value</b> |
|------------------------------------|--------------|
| ROC-AUC                            | 0.9911       |
| PR-AUC                             | 0.9990       |
| Test FPR (at 1% validation target) | 0.88%        |
| Attack Recall                      | 82.34%       |
| Attack Precision                   | 99.88%       |
| Attack F1                          | 0.9027       |

Table 4: Anomaly detection results (frozen validation-calibrated threshold). The detector achieves near-perfect precision (99.88%) at the cost of moderate recall (82.34%), a typical trade-off in security applications where false alarms are costly.

Figure 6 summarizes test-set performance:

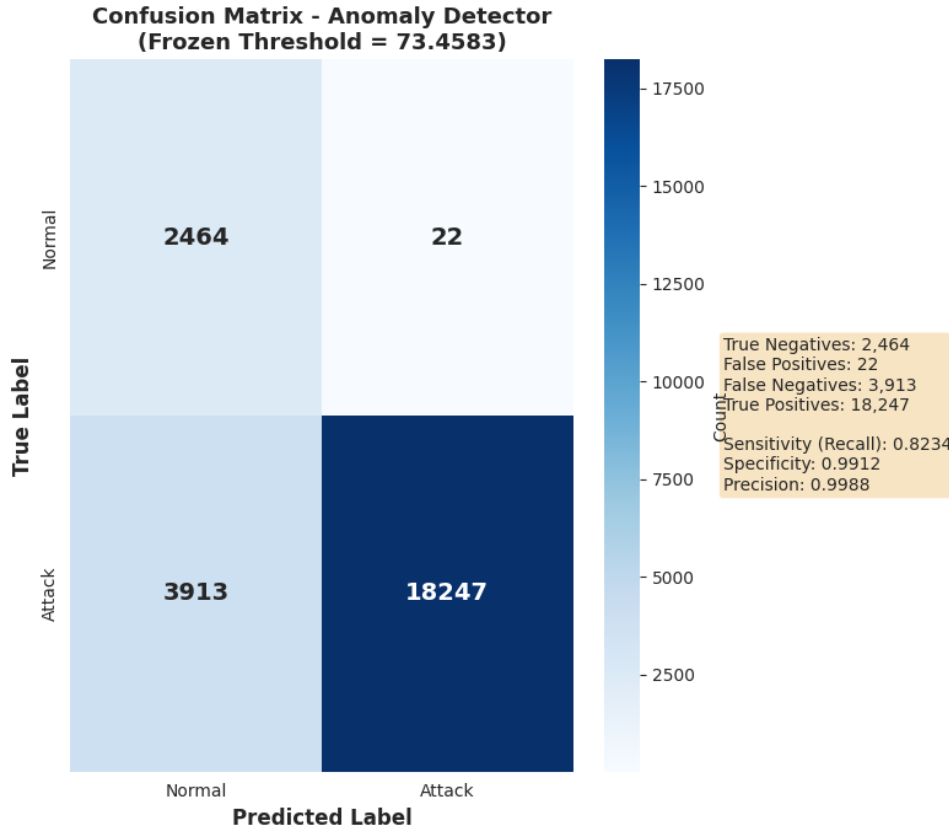


Figure 6: Confusion matrix for anomaly detection. True Negatives: 2,464 (normal flows correctly identified). False Positives: 22 (false alarms). True Positives: 18,247 (attacks correctly detected). False Negatives: 3,913 (missed attacks). Attack Recall (82.34%), Attack Precision (99.88%), and Attack F1 (0.9027) demonstrate excellent detection capability with minimal disruption to normal operations.

## 5 Discussion

### 5.1 Model-Dependent PCA Effects

The divergent response to PCA across models is noteworthy:

- **Logistic Regression:** Linear model most sensitive to dimensionality reduction. Reducing the feature space from 94 to 25 components (removing 69 dimensions, or 73.4% of the original dimensions) causes a 5.6% Macro-F1 loss, with the bootstrap CI providing evidence that the difference is systematic.
- **Random Forest:** Tree-based models are robust to feature subsets. RF loses only 0.7%, well within noise (CI contains zero). Random forests can work with any feature subset and often benefit from reduced overfitting, but here the gain is negligible.
- **XGBoost:** Gradient boosting actually improves 0.3% under PCA. This surprising result suggests PCA may reduce noise or resolve multicollinearity that XGBoost’s regularization cannot fully address.

**Implication:** Practitioners should not assume PCA uniformly helps or hurts. Model-specific validation is essential.

## 5.2 Class Imbalance Handling

Balanced class weights (`class_weight='balanced'` in LR/RF, sample weights in XGBoost) are applied during *training*, not evaluation. Evaluation uses Macro-F1 (unweighted average of per-class F1) and Balanced Accuracy (average recall per class). This separation is critical:

$$\text{Macro-F1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F1_c \quad (3)$$

where  $F1_c$  is the F1 score for class  $c$ . Even if a model trivially predicts all attacks (ignoring normal classes), Macro-F1 remains low because the normal-class F1 is zero.

## 5.3 Leakage Prevention via Grouping

The duplicate-feature grouping ensures that if a particular combination of features appears in both train and test, it stays on one side of the split. This is verified: `group overlap = 0`. However, this controls *feature-exact* leakage only. Session/device/time-window leakage is not explicitly evaluated; future work could implement temporal cross-validation.

## 5.4 Anomaly Detection Design

Training on normal traffic only ensures the model learns the normal distribution without attack contamination. The reconstruction-error anomaly detector assumes:

1. Attacks cause noticeable feature-space departure from normal.
2. PCA of normal data captures the normal manifold.
3. Out-of-distribution attacks incur higher reconstruction error.

In practice, this works well (ROC-AUC 0.9911), though the approach is less sensitive to adversarial attacks designed to evade the normal model.

## 6 Limitations

1. **Laboratory Data:** RT-IoT2022 is a testbed dataset under controlled conditions. Real-world IoT traffic may exhibit different distributions and attack patterns.
2. **Fixed Baseline Models:** No hyperparameter tuning. Optimized models would likely perform better but would introduce selection bias.
3. **Attack Prevalence:** Test-set attack prevalence is 89.9%, dominated by DOS\_SYN\_Hping. Real networks typically have much lower prevalence ( $< 1\%$ ). Our results do not directly translate to deployment.
4. **Unseen Attacks:** The anomaly detector is trained exclusively on normal traffic and does not explicitly learn attack-specific representations. Its ability to detect novel or zero-day attacks therefore depends on whether those attacks produce sufficiently large deviations from the learned normal feature-space structure.
5. **Reproducibility Note:** All random seeds are frozen. Results are deterministic given the dataset. However, the analysis does not claim generalization beyond RT-IoT2022.

## 7 Conclusions

This study demonstrates rigorous methodology for intrusion detection on severely imbalanced data:

- **Group-aware splitting** prevents exact-duplicate leakage (group overlap = 0).
- **Balanced evaluation metrics** (Macro-F1, Balanced Accuracy) ensure fair comparison despite 3,381:1 class imbalance.
- **Model-specific PCA analysis** shows that dimensionality reduction effects are heterogeneous: LR suffers 5.6% loss (significant), RF shows 0.7% loss (uncertain), XGB gains 0.3% (uncertain).
- **Bootstrap confidence intervals** quantify uncertainty in observed differences, distinguishing between random variation and systematic effects.
- **Normal-only anomaly detection** achieves ROC-AUC = 0.9911 and PR-AUC = 0.9990, demonstrating that unsupervised reconstruction-error methods can be highly effective on imbalanced data.

The contribution is *methodological*: correct handling of class imbalance, leakage prevention, and honest uncertainty quantification. While not state-of-the-art in detection performance, the analysis provides a template for rigorous evaluation of intrusion-detection systems on imbalanced datasets.